Predicting Nifty 50 Volatility Regimes Using Machine Learning

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Introduction

Financial markets exhibit varying levels of volatility, which significantly affect risk management and the performance of trading strategies. This project aims to analyse historical data of the **Nifty 50 (^NSEI)** index over a 15-year period to identify and predict "Volatility Regimes". By classifying market states as either High or Low volatility, investors can better tailor their strategies to current market conditions.

Problem Statement and Objectives

The primary challenge in market analysis is the volatility's non-stationarity. Predicting whether the market is entering a period of high turbulence or relative stability is crucial for:

- **Risk Mitigation**: Adjusting position sizes or hedging during high-volatility regimes.
- **Strategy Selection**: Switching between trend-following (low volatility) and mean-reversion (high volatility) strategies.
- **Machine Learning Integration**: Leveraging technical indicators and historical price patterns to provide a predictive framework for volatility.

Objective: To build a robust machine learning pipeline that downloads 15 years of Nifty 50 data, engineers a comprehensive set of predictive features, and evaluates multiple classification models to determine the most effective predictor of future volatility regimes.

Dataset and Variables

Data Source and Preparation

- **Source**: Historical daily data for the Nifty 50 (^NSEI) index (March 2011 to March 2026) was retrieved via the yfinance library.
- **Sample Size**: The final dataset consisted of **3,199 samples** after handling missing and infinite values.
- **Target Variable**: The volatility regime was defined based on the 20-day rolling volatility relative to its median. This target was shifted by one day to create a predictive model (predicting tomorrow's regime using today's data).

Feature Engineering (22 Total Features)

To capture diverse market dynamics, several categories of features were engineered:

- **Volatility Metrics**: 10-day, 20-day, and 30-day rolling volatility.
- **Momentum/Returns**: 5-day, 10-day, and 20-day daily returns.
- **Trend Indicators**: Moving Averages (MA 10D, 20D, 50D) and Price/MA ratios.
- **Technical Indicators**: RSI, MACD, Bollinger Band Width (BB Width), and Stochastic Oscillator.
- **Volume Indicators**: Volume change, Volume Moving Average, and Relative Volume.

- **Risk and Lagged Features**: Downside volatility, lagged returns, and lagged volatility.

Methodology

The machine learning pipeline followed a structured approach:

1. **Preprocessing**: Data was standardised using StandardScaler to ensure features were on a comparable scale.
2. **Model Selection**: Four models were trained and compared:
    - Logistic Regression
    - Random Forest Classifier
    - Gradient Boosting Classifier
    - XGBoost Classifier
3. **Train/Test Split**: An 80/20 time-series split (2,559 training samples, 640 testing samples) was utilised to preserve the temporal order of financial data.

Results and Model Performance

The models were evaluated using Accuracy, Precision, Recall, F1-Score, and ROC-AUC.

Model Comparison Table

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
XGBoost	0.9578	0.9419	0.8982	0.9179	0.9854
Logistic Regression	0.925	0.9685	0.7365	0.8367	0.9829
Random Forest	0.9547	0.9313	0.8922	0.9113	0.9787
Gradient Boosting	0.9359	0.8462	0.9222	0.8825	0.9799

Key Evaluation Findings

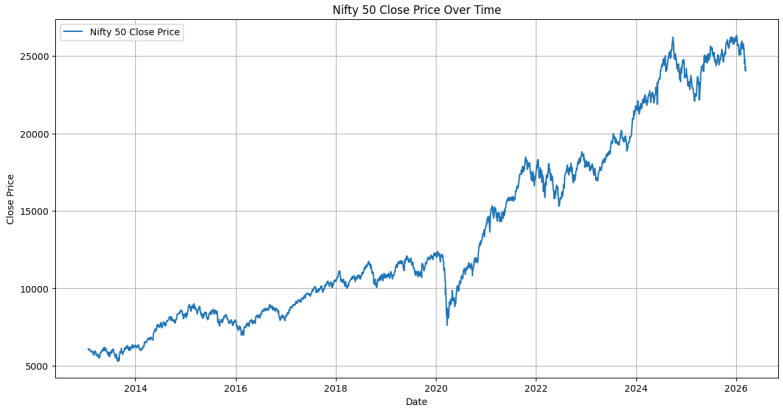
- **Best Model**: **XGBoost** was identified as the best-performing model with the highest ROC-AUC (0.9854) and F1-Score (0.9179), indicating an exceptional balance between precision and recall.
- **Logistic Regression**: Showed high precision (0.9685) but lower recall (0.7365), suggesting it was conservative in flagging high-volatility regimes.
- **Ensemble Power**: The tree-based ensemble models (XGBoost and Random Forest) significantly outperformed linear methods in capturing complex market patterns.

Key Observations and Visualisations

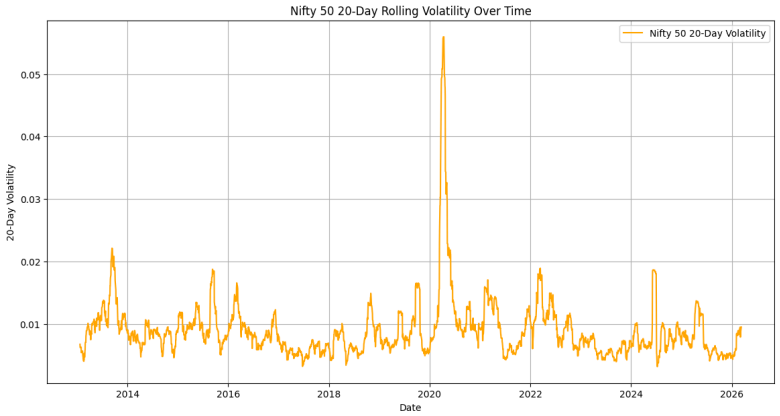
Visualizations Used

The project utilised several key visual tools to interpret findings:

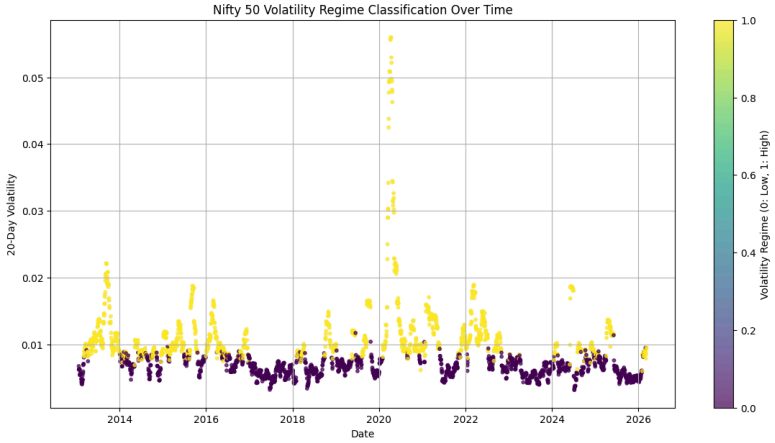
- **Nifty 50 Close Price Over Time**: Provided a baseline for understanding long-term index growth and major market drawdowns.



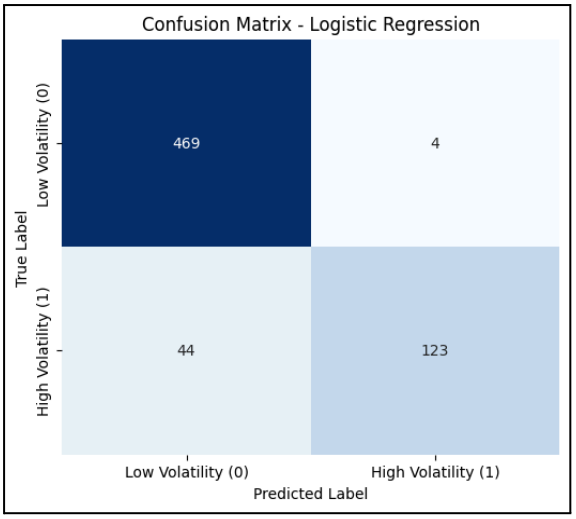
- **Nifty 50 20-Day Rolling Volatility Over Time**: Highlighted periods of "volatility clustering" where high volatility followed high volatility.

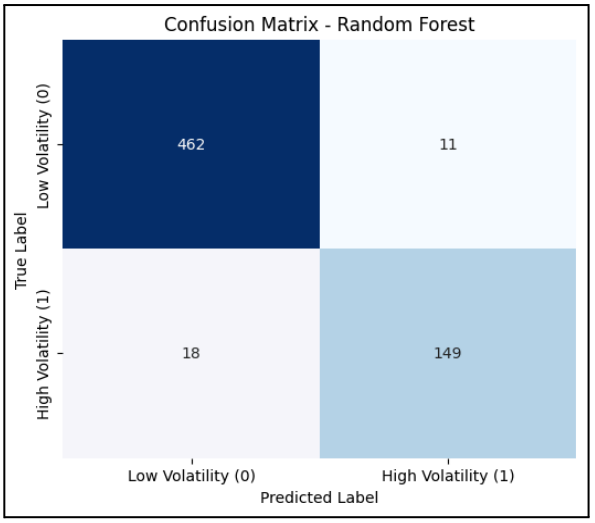


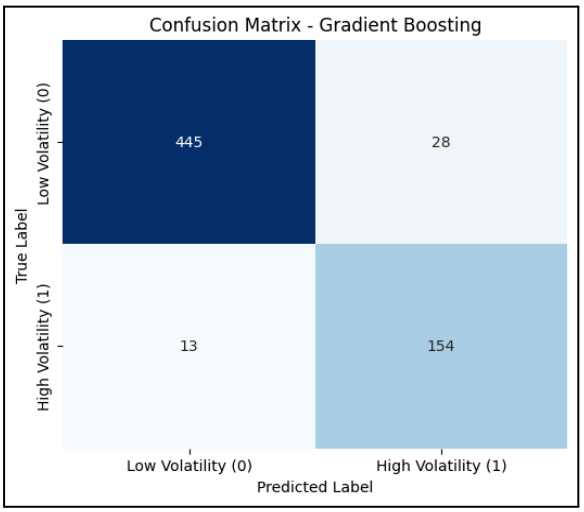
- **Volatility Regime Classification Over Time (Scatter Plot)**: Successfully mapped historical dates to High/Low regimes, showing clear separation during market stress.

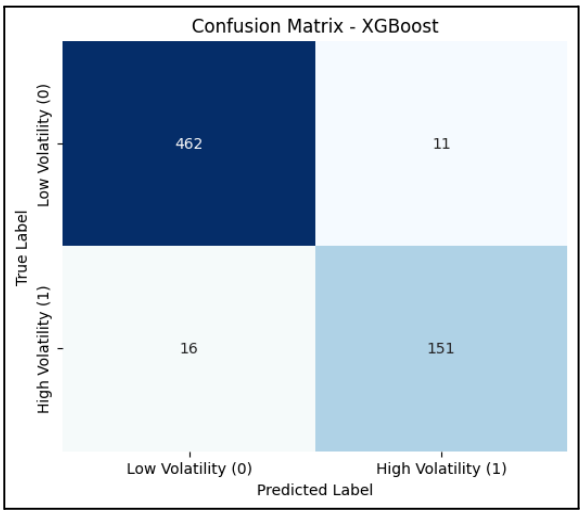


- **Confusion Matrix**: For the XGBoost model, the matrix confirmed a high rate of True Positives and True Negatives with minimal misclassifications.

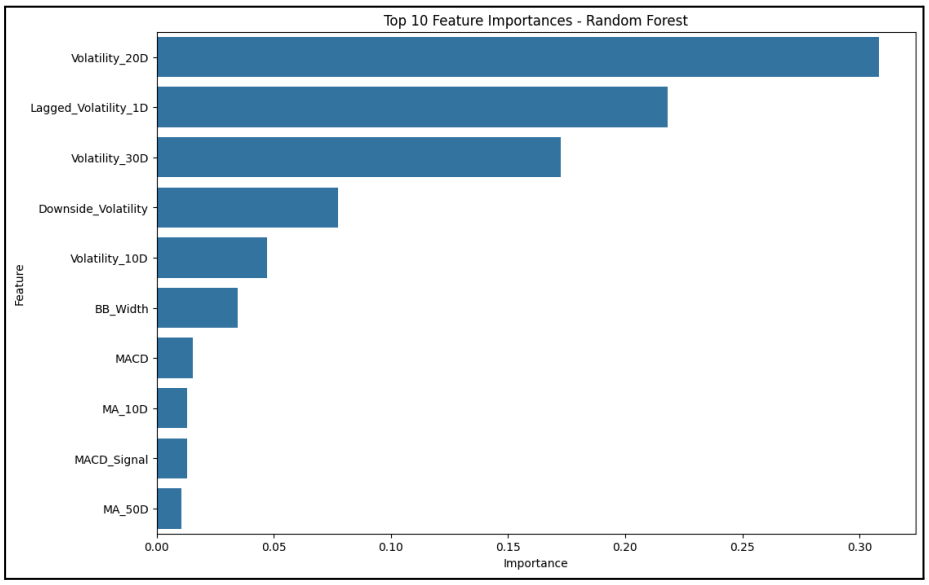


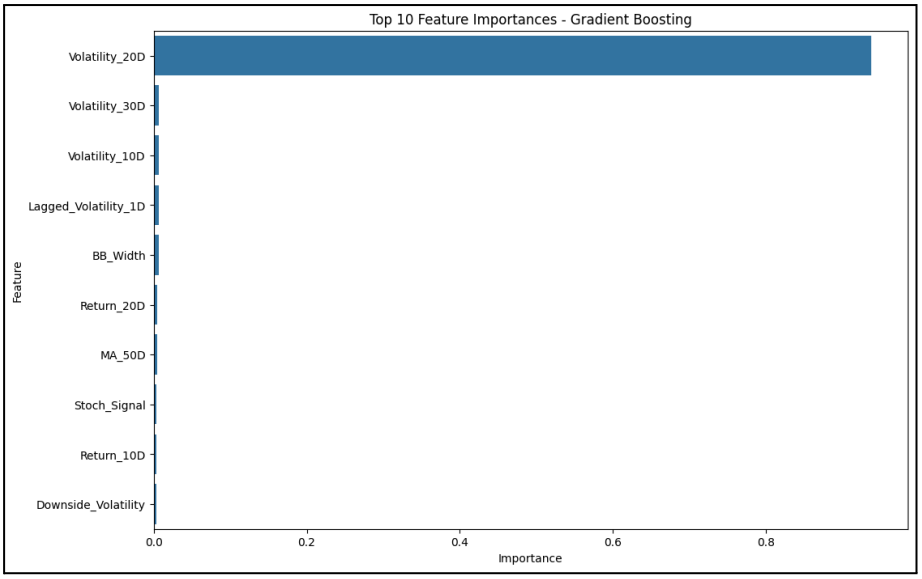


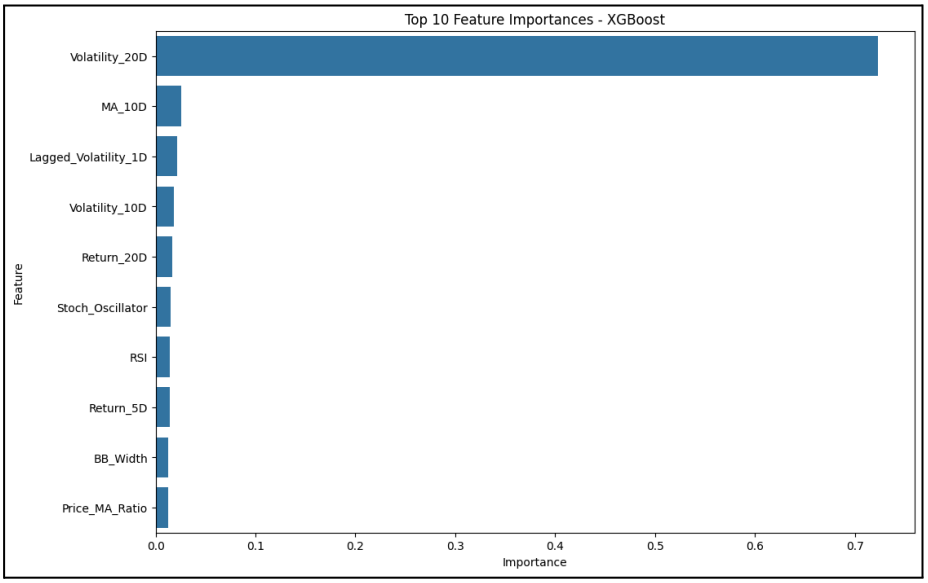




- **Feature Importances for each model**







Top 10 Feature Importances of XGBoost

The model highlighted that volatility and trend are the primary drivers of future regimes:

1. **Volatility_20D** (Highest Importance)
2. **MA_10D**
3. **Lagged_Volatility_1D**
4. **Volatility_10D**
5. **Return_20D**
6. Stochastic Oscillator
7. RSI
8. Return_5D
9. BB Width
10. Price_MA_Ratio

Conclusion and Practical Implications

The study concludes that Nifty 50 volatility regimes are highly predictable using past price movements, technical indicators, and volatility metrics. The XGBoost model demonstrated near-perfect discriminatory power (ROC-AUC ~0.985) in identifying market states.

Practical Usage

- **Algorithmic Trading**: Traders can use this model to dynamically toggle between aggressive and defensive strategies based on the predicted regime for the next day.
- **Options Pricing**: Since option premiums are driven by implied volatility, predicting shifts in regime can help identify overvalued or undervalued options.
- **Institutional Risk Management**: Portfolio managers can implement "volatility targeting" by scaling down exposure when the model predicts a shift from a low to a high-volatility regime.

Future Work: Incorporating macroeconomic factors (e.g., interest rates, inflation) and sentiment analysis from financial news could further enhance the model's predictive robustness.